

Program : Diploma in Communication and Computer Networking	
Course Code: 4327	Course Title: Microcontrollers Lab
Semester : 4	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3(L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Implement simple programs using microcontroller.
- Implement interrupt handling programs
- Perform interfacing experiments with 8051.

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive Level
CO1	To write simple programming experiments with 8051	9	Applying
CO2	To prepare branching and control instructions with 8051	8	Applying
CO3	To practice interrupt handling programs	6	Applying
CO4	To demonstrate interfacing 8051 with peripherals	14	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2	3			
CO2	3		2	3			
CO3	3		2	3			
CO4	3		2	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To write simple programming experiments with 8051		
M1.01	Implement assembly language programs based on Data transfer instructions	3	Applying
M1.02	Implement assembly language programs for Arithmetic instructions (8/16 bit addition, subtraction, multiplication and division)	3	Applying
M1.03	Implement assembly language programs for logical instructions (AND, OR etc)	3	Applying
CO2	To prepare branching and control instructions with 8051		
M2.01	Implement assembly language programs for Branch instructions	2	Applying
M2.02	Implement assembly language programs for looping concepts	2	Applying
M2.03	Implement assembly language programs for counting concepts	2	Applying
M2.04	Implement assembly language programs to introduce delay using Timer / Counter	2	Applying
	Series Test– I	2	
CO3	To practice interrupt handling programs		
M3.01	Develop Assembly Language Programs for Interrupts	6	Applying
CO4	To demonstrate interfacing of 8051 with peripherals		
M4.01	Implement programs for serial communication	2	Applying
M4.02	Develop a program to interface LED with 8051	2	Applying
M4.03	Develop a program to interface 7 segment Display with 8051	2	Applying
M4.04	Develop a program to Interface 8 bit DAC with 8051	2	Applying

M4.05	Develop a program to interface a DC Motor with 8051	2	Applying
M4.06	Develop a program to interface a Stepper Motor with 8051	2	Applying
M4.07	Develop a program to interface a Stepper Motor with 8051	2	Applying
	Series Test -II	2	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.

Minimum of 6 experiments (excluding open ended experiments/microprojects) shall be performed.

Compulsory for CIA

The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
T1	Raj Kamal, <i>Microcontrollers: Architecture, Programming, Interfacing and System Design</i> , Pearson Education, 2011.
T2	Kenneth J. Ayala, <i>The 8051 Microcontroller</i> , Cengage learning, 3/e.
R1	Han Way Hung, "PIC Microcontroller, An introduction to software and hardware interfacing", Cengage learning
R2	I.Scott Mackenzie, Raphael C.-W Phan, <i>The 8051 microcontroller</i> , 4th edition.

Online Resources

Sl No	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee18
2	https://study.unisa.edu.au/courses/151843